

GATE CSE 2026

Predicted Paper

GATE CS 2026 - Predicted Paper 4: High-Yield Integration

Focus: Programming and Data Structures, Discrete Mathematics, Compiler Design, DBMS, OS, CN

Total Questions: 65 | Total Marks: 100 | Duration: 180 min

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General Aptitude

Q1. [1 mark] Synonym of: MITIGATE

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q2. [1 mark] A does work in 20 days, B in 30. Together?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q3. [2 marks] Find error: The scissor is kept on the table.

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q4. [1 mark] $\log_{10} = ?$

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q5. [2 marks] CP = Rs 500, Profit 20%. SP = ?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q6. [1 mark] OPPOSITE of: TRANSIENT

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q7. [1 mark] If $a:b = 2:3$, $b:c = 4:5$, then $a:c = ?$

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q8. [2 marks] Choose synonym: LOQUACIOUS

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q9. [1 mark] Area of circle with $r=7$ ($\pi=22/7$)?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Q10. [2 marks] A is $2x$ B. B is $3x$ C. $A:C = ?$

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Engineering Mathematics

Q11. [1 mark] Trace of $\begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$?

- @. [object Object]
A. [object Object]
B. [object Object]
C. [object Object]

Explanation: Trace = sum of diagonal elements = 3+4=7.

Q12. [1 mark] Variance of uniform distribution $[a,b]$?

- @. [object Object]
A. [object Object]
B. [object Object]
C. [object Object]

Explanation: $Var(U[a,b]) = (b-a)^2/12$. Mean = $(a+b)/2$.

Q13. [1 mark] integral of $2x \, dx = ?$

- @. [object Object]
A. [object Object]
B. [object Object]
C. [object Object]

Explanation: Power rule: $\int x^n dx = x^{n+1}/(n+1)$. $\int 2x dx = x^2 + C$.

Q14. [2 marks] Relation $R=\{(1,1),(2,2),(3,3)\}$ on $\{1,2,3\}$ is:

- @. [object Object]
A. [object Object]
B. [object Object]
C. [object Object]

Explanation: All pairs (x,x) present $\rightarrow$ reflexive. Symmetric and transitive trivially.

Q15. [2 marks] $A^T A$ is always:

- @. [object Object]
A. [object Object]
B. [object Object]
C. [object Object]

Explanation: $(A^T A)^T = A^T (A^T)^T = A^T A$. Always symmetric and positive semi-definite.

Technical (CSE)

Q16. [1 mark] Full binary tree with n internal nodes $\rightarrow$ leaves:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Full binary tree: n internal nodes $\rightarrow n+1$ leaves. Total nodes = $2n+1$.

Q17. [1 mark] Circular linked list last node points to:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Circular LL: last.next = head. No NULL termination.

Q18. [1 mark] Stack is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Stack = Last In First Out (LIFO).

Q19. [1 mark] Queue is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Queue = First In First Out (FIFO).

Q20. [2 marks] Inorder traversal of BST gives:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Inorder traversal of BST yields elements in ascending (sorted) order.

Q21. [2 marks] Binary tree height h has max nodes:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Perfect binary tree at height h has $2^{(h+1)}-1$ nodes (if h =edges) or 2^h-1 (if h =levels).

Q22. [2 marks] Min-heap property:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Min-heap: parent value $\leq$ child values. Max-heap: opposite.

Q23. [2 marks] Load factor $\alpha = n/m$. $n=50$, $m=25$. $\alpha = ?$

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: $\alpha = n/m = 50/25 = 2$.

Q24. [1 mark] Height of AVL tree with n nodes is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: AVL tree is height-balanced $\rightarrow$ height = $O(\log n)$.

Q25. [1 mark] BFS uses:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: BFS uses queue (FIFO). DFS uses stack (LIFO).

Q26. [2 marks] Recursion needs:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Every recursive function needs: base case (termination) + recursive case.

Q27. [1 mark] Collision resolution by chaining uses:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Chaining stores colliding elements in linked list at each hash slot.

Q28. [2 marks] Postorder of expression tree gives:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Postorder traversal = postfix (Reverse Polish) notation.

Q29. [2 marks] Stack applications EXCEPT:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: BFS uses queue, not stack. Stacks used for: parenthesis, recursion, DFS, expression eval, function calls.

Q30. [2 marks] Max nodes at level l (root=level 0)?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: At level l, maximum = 2^l nodes. Full binary tree doubles each level.

Q31. [1 mark] A union empty set = ?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Union with empty set: $A \cup \{\} = A$.

Q32. [1 mark] Relation that is reflexive, symmetric, transitive:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Equivalence relation = reflexive + symmetric + transitive.

Q33. [1 mark] $f:A \rightarrow B$, every element of A maps to unique B:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Definition of function: each input has exactly one output.

Q34. [2 marks] $p \rightarrow q$ is equivalent to:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Implication: $p \rightarrow q = \text{not } p \text{ OR } q$. Only false when $p=\text{true}$, $q=\text{false}$.

Q35. [2 marks] Every bounded lattice has:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Bounded lattice = has both greatest upper bound (1) and least lower bound (0).

Q36. [2 marks] Group with commutative operation:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Abelian group = group + commutative operation.

Q37. [2 marks] $C(n,k)$ where $n=10$, $k=3$ = ?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

*Explanation: $C(10,3) = 10!/(3!*7!) = (10*9*8)/(3*2*1) = 720/6 = 120$.*

Q38. [1 mark] Eulerian circuit requires:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Eulerian circuit: all vertices have even degree.

Q39. [1 mark] Lexeme -> ? -> Token

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Lexeme matched against Pattern to produce Token.

Q40. [2 marks] LR parser type:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: LR = Left-to-right, Rightmost derivation in reverse -> bottom-up.

Q41. [2 marks] Backpatching used in:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Backpatching fills jump addresses for control flow in intermediate code.

Q42. [1 mark] Three-address code example:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: TAC uses temporaries: $T1 = b + c$; $a = T1$.

Q43. [2 marks] LALR parser:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: LALR = Look-Ahead LR. Merges LR(1) states with same cores.

Q44. [1 mark] BCNF is stricter than:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: BCNF is stricter than 3NF. BCNF subset of 3NF.

Q45. [2 marks] ACID 'A' stands for:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: A=Atomicity, C=Consistency, I=Isolation, D=Durability.

Q46. [2 marks] UNIQUE constraint allows:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: UNIQUE allows multiple NULLs (NULL != NULL in SQL).

Q47. [2 marks] B+ tree leaf nodes:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: B+ tree: leaves store keys + data pointers. Internal nodes store keys + child pointers.

Q48. [1 mark] Two-phase locking:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: 2PL ensures conflict-serializable schedules. Does not prevent deadlocks.

Q49. [1 mark] SJF is:

- @. [object Object]
- A. [object Object]

- B. [object Object]
- C. [object Object]

Explanation: SJF can be non-preemptive or preemptive (SRTF = Shortest Remaining Time First).

Q50. [2 marks] Page fault service time (no TLB, page not in memory)?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Page fault: fetch page from disk -> load into memory -> restart. Dominated by disk I/O.

Q51. [2 marks] Producer-consumer uses:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Producer-consumer uses counting semaphores (empty/full) + mutex.

Q52. [1 mark] Four necessary conditions for deadlock:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: All 4 required: mutual exclusion, hold & wait, no preemption, circular wait.

Q53. [1 mark] Process Control Block contains:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: PCB stores process state, registers, memory maps, scheduling info, etc.

Q54. [1 mark] IP address class B first octet range:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Class B: 128-191. Class A: 0-127, C: 192-223, D: 224-239.

Q55. [2 marks] Subnets from /24 using /26 mask?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: /26 from /24: $2^{(26-24)} = 2^2 = 4$ subnets.

Q56. [2 marks] TCP congestion window on timeout:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: On timeout: $ssthresh = cwnd/2$, $cwnd = 1$, restart slow start.

Q57. [1 mark] SMTP is for:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: SMTP = Simple Mail Transfer Protocol for email.

Q58. [2 marks] CRC polynomial x^3+x+1 . Message 1010. CRC bits?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Generator degree 3 -> CRC = 3 bits appended to message.

Q59. [1 mark] Merge sort is:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Merge sort = stable (preserves order of equal elements), not in-place (needs $O(n)$ extra).

Q60. [2 marks] $T(n)=3T(n/2)+O(1)$. Master theorem:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: $a=3$, $b=2$. $n^{(\log_2 3)}$ approx $n^{1.58}$. $f(n)=O(1) = O(n^{(\log_2 3 - \epsilon)})$. Case 1 $\rightarrow O(n^{\log_2 3})$.

Q61. [2 marks] Activity selection (interval scheduling) uses:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Activity selection is greedy: sort by finish time, select compatible activities.

Q62. [2 marks] Strongly connected components (Kosaraju):

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Kosaraju: DFS on original $\rightarrow$ DFS on transpose in order of finishing times. Tarjan uses single DFS.

Q63. [2 marks] Pipeline stall due to data hazard:

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: Forwarding reduces stalls but RAW hazards with load-use still cause 1 stall.

Q64. [1 mark] IEEE 754 single precision mantissa bits?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: IEEE 754 single: 1 sign + 8 exponent + 23 mantissa = 32 bits.

Q65. [2 marks] Hit ratio 95%, access 2ns (hit), 20ns (miss). EAT?

- @. [object Object]
- A. [object Object]
- B. [object Object]
- C. [object Object]

Explanation: $EAT = 0.95 \cdot 2 + 0.05 \cdot 20 = 1.9 + 1.0 = 2.9ns$.

Answer Key

General Aptitude

Q1: B
Q2: B
Q3: C
Q4: B
Q5: B
Q6: B
Q7: A
Q8: B
Q9: A
Q10: B

Engineering Mathematics

Q11: C
Q12: A
Q13: C
Q14: D
Q15: A

Technical (CSE)

Q16: B
Q17: B
Q18: B
Q19: B
Q20: B
Q21: C
Q22: B
Q23: B
Q24: B
Q25: B
Q26: B
Q27: B
Q28: B
Q29: C
Q30: B
Q31: B
Q32: A
Q33: B
Q34: A
Q35: B
Q36: B
Q37: A
Q38: B
Q39: A
Q40: B
Q41: B
Q42: B
Q43: B
Q44: C
Q45: B
Q46: B
Q47: B
Q48: B
Q49: B
Q50: B
Q51: B
Q52: B
Q53: B
Q54: B
Q55: B
Q56: B
Q57: B
Q58: C
Q59: B
Q60: B
Q61: B
Q62: B

Q63: B
Q64: A
Q65: A